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* An experiment is a process whose outcome
s not known with certainty.

- The set of all possible outcomes of an experiment
s called sample space (denoted by S)

- The outcomes themselves are called sample points
n the sample space.

$$S = \{H, T\}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

Sample points $\rightarrow$ Random variable.

Range of Random Variable : $-\infty$ to $+\infty$

$$S = \{1.7, 1.8, \ldots, 2.0\}$$

* Discrete : A random variable X is said to
be discrete if it can take on a countable
number of values.

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- The probability that X takes on the value
x_i is given by:

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2. $F(x)$ is nondecreasing.

if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$

3. $\lim_{x \to \infty} F(x) = 1$ and $\lim_{x \to -\infty} F(x) = 0$

since X takes only finite values

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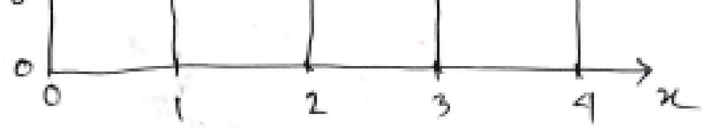
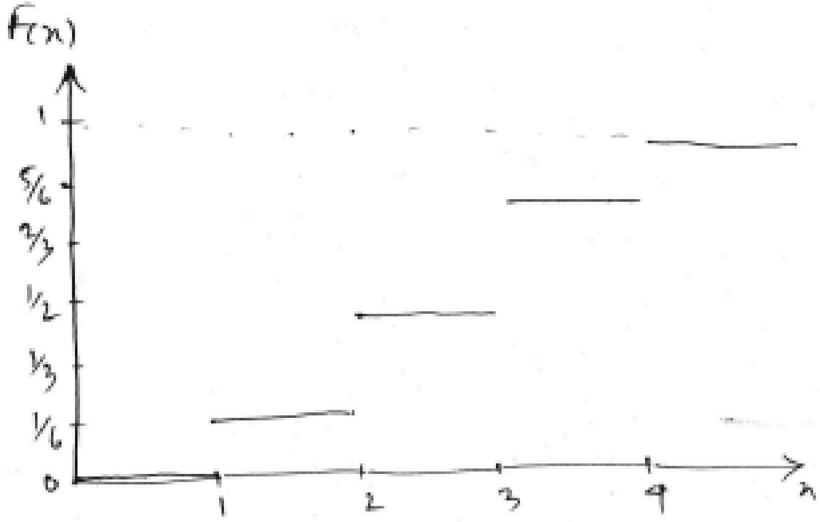


Probability distribution function:



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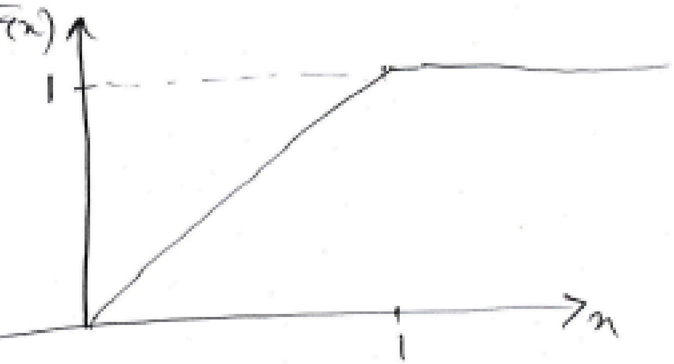
$$= \lim_{n \to \infty} \sum_{i=1}^{n} 1 \cdot \Delta y \qquad \Delta y = \frac{x-0}{n} = \frac{x}{n}$$

$$\sum_{i=1}^{n} 1 = n$$

$$= \lim_{n \to \infty} \cancel{n} \cdot \frac{x}{\cancel{n}}$$

$$= x$$



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$(y)\, dy = \int_x 1 . dy$

$\sum_{k=1}^{n} 1 . \Delta y \qquad \Delta y = \frac{x + \Delta x - x}{n} = \frac{\Delta x}{n}$

$n . \frac{\Delta x}{n} \qquad \sum_{k=1}^{n} 1 = n$

x

t a uniform random variable
likely to fall in any interval of
en 0 and 1, which justifies the
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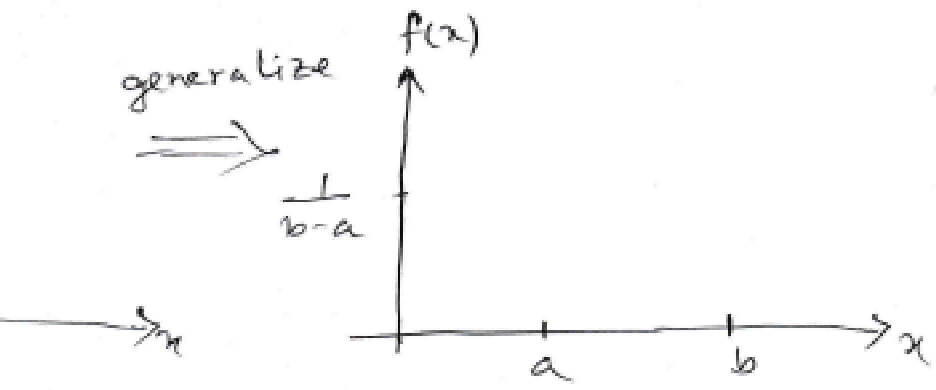
ution : has constant probability
domain.



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$) = 1$

$= \text{height} \times \text{width}$

$= \text{height} \times (b-a)$

$\text{ght} \times (b-a) = 1$

$\text{ight} = \frac{1}{b-a}$

Range: $[a, b]$

Mean (μ): $\frac{b+a}{2}$

Variance : $\frac{(b-a)^2}{12}$

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